ActInf Livestream #051.1 ~ "Canonical neural networks perform active infe

Livestream | 1:58:03

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

Thank you.

active inference from a variety of different perspectives, from the more fundamental math
and physics to some applications, philosophy, embodiment, all these really interesting threads.
And this paper seems to make a really clear, meaningful contribution and connection by
connecting active inference entities and this approach of modeling to neural networks, which
are in daily use globally.
So thought it was a fascinating connection and really appreciate that we can talk about
this today.
So to you and welcome.
Go for it Takuya, however you'd like to introduce and say hello.
Oh, yeah.
Hi, I'm Takuya Isomura, a neuroscientist in Rikan Brain Science Institute in Japan.
I'm particularly interested in universal characterization of neural network and brain using mathematical
techniques.
So this work, I believe, is important as a link between active inference aspect, vaginal inference
aspect of the brain and the dynamic curve system aspect of the neural network.
So I'm very happy to join this discussion session.
Thank you for the invitation.
Nice to meet you.
Nice to meet you as well.
The first thing you added, the universal characterization of neural networks.
What is the universal characterization of neural networks?
Why is it being pursued in this area of research?
So as a narrow sense, my main aim of this paper is that people use active inference formalization
to characterize brain activity behavior, so on, so on, but which would be different from a conventional
neural network.
So there is a class of problem which is associated with conventional neural network.

And it is not very clear whether all characterization of conventional neural network can be explained

by active inference free energy principle or not.

So here universal characterization means that characterization of every aspect of conventional neural network,

which is a kind of dynamical system derived as the association between a physiological phenomenon

and simple mathematical formula using typical using differential equations.

As the broad sense, I think universal characterization means that, well, it is the characterization of brain intelligence,

but it is a big picture and what the paper particularly addresses only one aspect of the full picture.

All right.

So it'll be great to pull back to really understand what synthesis is happening.

So I'm going to ask what makes a neural network model, a neural network model, and what makes an active inference model, an active inference model?

Is this synthesis and connection you've made true because of what?

Because of, you know, basically what we showed is the, the mathematical equivalence between the formulation and the

the canonical neural networks and the formulation of active inference.

In the sense that we showed that a class of neural network can be characterized by a minimization of some

biological plausible cost function.

And we show that that function can be read as a class of variation of Bayesian inference and a particular class of general inference.

In terms of, well known partially of the bubble neural constitution process.

All right.

Shall we perhaps walk through some of the sections of the paper?

It would be awesome just for each of these sections, maybe the numbered and the lettered sections.

What does the section aim to show and why was it there in the paper?

So, uh, yeah, briefly.

So first we introduce, uh, so the main main issue, main problem, our interest, which is a relationship.

So which is that, so we, we try to make a formal link between neural network and active reference.

That's a, that's the main problem, uh, background.

And then we first formulate, formulate, uh, the equivalence, mathematical equivalence in a very broad manner.

So in the, uh, uh, uh, the, the relationship using, uh, uh, complete class theorem, which is, uh, well known, uh, statistical theorem, uh, proposed very long time ago.

And, uh, uh, we, using that, we, we link, uh, a general form of neural network with, uh, a general form of, uh, variational Bayesian inference.

But, uh, uh, uh, uh, program is that, uh, this characterization does not addresses a specific hormone generative model, uh, which is, uh, crucial to characterize, you know, uh, a specific, specific model, specific neural network dynamics.

So, uh, uh, in the following, uh, uh, sections, we, uh, characterize, uh, the problem using POMDP, or partially observable marker digital process.

And, uh, link, uh, that, uh, model with a particular class of, uh, canonical neural network.

And then, uh, we simulated, uh, we, we, we use the simulation to, to, uh, uh,

uh, correlate, uh, that property, uh, in terms of some maze tasks.

All right.

Thank you for this.

Could we talk about the complete class theorem?

So, what is the scope of the complete class theorem?

And why was it the relevant set of the neural networks to pursue or the right way to frame it?

Yeah.

Thank you for asking that.

So I like, uh, the, the slide you showed the last week's, uh, video.

So, uh, complete the set cross theorem basically indicate that relationship between, uh, some class of decision rule and Bayesian inference.

Uh, here, a crucial keyword is, uh, uh, uh, admissibility or admissible decision rule, which is a rule, uh, which is, uh, as same as, uh, as, as good as, uh, other, uh, decision rules or, uh, at least in, uh, at one point, uh, better than other decision rules.

So, yeah, simply speaking, uh, admissibility indicate, uh, in some sense, it is a best, uh, rule, uh, for some aspect.

And usually we characterize such a goodness using, uh, cost function, loss function, or risk function.

And here, what we, uh, did is, uh, some, we, uh, established some association with this, this type of loss function or risk function with, uh, uh, function of canonical neural network, uh, which is, we call cost function or biologically plausible cost function for neural network.

So, uh, so our assumption is that, uh, uh, neural network minimized cost function.

So if it achieved the minimization, and it is, uh, obviously, uh, achieved some sort of optimality.

So we can say it is admissible with respect to that cost function.

So the beauty of, uh, complete cross theorem is that, uh, if, uh, we, we find, uh, some admissible decision rules, then automatically we can say that it is based based on inference in terms of, in terms of some Bayesian cost function with, uh, general model or prior beliefs.

So, uh, uh, this, uh, complete cross theorem is a crucial as a, you know, abstract, uh, characterization of the relationship between conventional neural network architecture dynamics and, uh, variational Bayesian inference.

Right.

Thank you.

Um, what does it mean when you said it was biologically plausible of a loss function?

Uh, with the term is a little bit arbitrary because so in this paper, we, uh, mean by, uh, biological plausibility, uh, by, you know, the, in the sense that, uh, this neural network is a, uh,

uh, this neural network model, uh, can be derived from realistic neural model, uh, through some approximation.

And, uh, uh, so here, uh, biologically plausibility, uh, suggests or means, you know, uh, plausibility as a neural model or synaptic plus system model.

Uh, and, uh, if, if this cost function loss function can derive, uh, such a plausible algorithm, then, uh, we can say that, uh, this cost function is very plausible.

So what is the distinction between those neural and synaptic components in the loss function or what equation to look at?

Uh, you mean distinction between neural dynamics and synaptic?

Yeah.

What is the distinction between them and how is it represented in the equations?

Uh, okay.

Uh, basically, uh, neural activity equation means, uh, uh, differential equation about a variable, uh, that represent, uh, firing intensity or, uh, some sort of, uh, variables associated with the, uh, neural, neural firing.

On the other hand, uh, synaptic elasticity equation means, uh, an update rule about the synaptic weights or synaptic strengths, uh, which is, uh, a connection between two neurons.

And, uh, uh, beauty of, uh, this formulation, uh, proposed in this paper is that, uh, we characterize both, uh, neural activity equation and synaptic

and synaptic velocity equations in terms of, uh, uh, well, uh, gradient descent on a same cost function, uh, common cost function.

So, uh, we can, uh, we can, uh, we can say that if we, uh, consider the, uh, partial derivative of some cost function with respect to neural activity, then it derives, uh, gradient descent rule about, uh, activity, neural activity.

While if we consider a partial derivative of cost function L with respect to synaptic weight, then we derive a synaptic plastic rule.

Are those the only two aspects of a neural network or why are those the two key aspects?

Uh, it is a main, uh, I, I think it's, uh, it's, uh, it's, uh, it's a main, uh, body of the neural activity.

If we, we, we consider some, uh, uh, inference running or action, uh, exhibit by neural network in the sense that, uh, neural activity correspond to fast dynamics, fast gradient, uh, dynamic.

And, uh, while, uh, synaptic plasticity indicate a slow dynamics that minimize, uh, risk function, cost function.

But in general, we can consider any, any aspect, any variables associated with neural network.

For example, uh, at least what we show in the paper is, uh, any free parameter, uh, which may be associated with, uh, firing threshold.

Or, uh, although we don't discuss in this paper, it would be possible to add other, uh, variables related to neural network.

For example, we here, we ignore, uh, contribution of Grier factor, uh, but, uh, it will be possible to add the Grier factor in this formulation or any other aspect of biological neural network.

Hmm.

That's very interesting.

And it speaks also to a general, uh, general separation of timescales.

timescales. For example, in different multiscale systems or in the renormalization group,

where it's describing some minimal multi-timescale system where the faster timescale can be seen as perception-like, and the slower timescale can be seen as more learning-like. And then in some

hierarchical model, what's learning of one timescale can be perceptual for a slower timescale.

So it's a very nice generalization.

Are there any examples of decision rules that will help us think about the action

components of what the neural network is doing? Because it may be more familiar to think about

digit characterization and image classification, some kind of classical tasks for neural networks.

But how does the decision rule play out in the context of neural networks?

Okay. So in this paper, we basically assume a closed loop, so comprising a neural network part and

the environmental part. So neural receives sensory input from environment and

the feedback, provide some feedback to the environment. So in, yeah, even with the example of digit

digit classification, we can say that output corresponds to a classification output, which is a kind of decision rule.

More relevant example would be, for example, controlling agent like a robot control or any kind of controlling or decision-making tasks.

For example, when we encounter some choice task, we need to output, for example, left or right or something.

Any kind of such decisions can be associated with the admissibility or admissible decision rule.

So what would an example of an admissible strategy be in the decision-making task?

Admisibility usually characterized by loss function or risk function. So if so here in admissible

in admissible strategy, we indicate that there is another decision rule, which is at least one point better than the forecast decision rule.

So yeah, simply speaking in admissible strategy indicate that it's that decision rule is not better, not good.

But let's just say our decision rule is we always turn right.

Is that an example of a decision rule? Because there might be settings where that is strictly effective and the simplest rule.

Whereas there's other settings where that's going to be tragic.

So what does it mean to be admissible for an agent in light of different environmental contexts?

That's an interesting point. So even with such a too much simplified rule, it can be admissible under some particular situation, particular loss function.

For example, the rule that always turn right, maybe, maybe the best under some situation, right?

So the relationship of admissibility or inadmissibility depends on both agent characteristic and environmental characteristic.

What aspects of the environment?

What aspects of the environment?

For example, if the decision rule matches the structure, architecture of environment, then, you know, maybe that decision always turn right, achieve the shortest path.

Right?

Under some situation, some environment.

How does this admissibility help us think about like overfitting?

And how does it help us think about the way that different practices are used for neural networks to prevent them from being overfit in practice?

Well, strictly admissibility is characterized with the Bayesian risk.

So we cannot observe a hidden state of the environment.

Only we can observe a part of the entire universe.

So the question is, what is the best choice under such a limited, limited information, limited information?

So this Bayesian risk idea or admissibility or complete clear scenario tell us that, so, you know, well known Bayesian, only the well known Bayesian framework achieved the admissive, admissibility, admissible,

which means that in this aspect, Bayesian optimization is give us a best best choice strategy.

Otherwise, we overfit or find the, you know, suboptimal solution.

So it's a nice, nice association, nice linkage between the decision.

What is a good decision about the decision?

And the more established statistical influence Bayesian influence framework.

Thank you.

Thank you.

That's very helpful.

So we're reducing our uncertainty and risk about hidden states in the environment.

So in the special case where the entire environment is observable without error, like a chess game, then there's an equivalence between calculation of risk or loss on observables or on hidden states, but they're not really hidden, but they are environmental states.

Whereas any amount of uncertainty in the mapping between observations and hidden states, which is usually shown as a in the partially observable Markov decision process, any amount of uncertainty about unobserved or partially observed environmental states.

Enables you to fit your uncertainty optimally about that hidden state.

And fit that uncertainty simply with a gradient descent.

And by doing so, you don't overfit a model of observables, which might be the fallacy or the issue with simply doing descriptive statistics.

You might get an infinitely small variance with a frequentist estimate because you have 100,000 data points.

So the variance from a descriptive statistics perspective might be very small.

I think it speaks very much to why neural networks are useful in practice from training with limited data sets.

Because that's an empirical observation that they don't entirely overfit.

But also I'm sure there's ways to construct them that are overfit.

Yeah.

Overfit would occur if we select some optimal prior beliefs, for example.

Well, I'm not sure if it's either overfit in the sense what you mentioned.

Because if we select some prior beliefs, then the Bayesian function itself changes.

And the neural networks try to fit to that cost function.

So cost function minimization would be achieved in such a situation.

But that solution is not good for our original purpose.

That's a tricky part.

Yeah.

That is reminiscent of some discussions we've had discussing like driving off a cliff or blowing up is also reducing free energy.

Like dropping off a building reduces your potential energy.

And so there are potentially decision making or strategic trajectories that do for some time horizon minimize free energy.

Yeah.

Perhaps even or maybe even guaranteed better than some longer time horizon.

Because if the short term strategies were somehow better than the long term horizon, it would be difficult to imagine.

Because the long term horizon would be at least as good as a short term strategy.

So that speaks to the challenges of planning in action.

So how is planning addressed in modern neural networks?

And how does this work help us think about that?

That's another very important aspect.

So first I have to say that this framework addresses planning aspect.

But that planning is not necessarily the optimal solution.

In the sense that what we are interested in is optimization or planning under a limited structure.

The structure is characterized by here,

vertically plausible two layer neural networks.

So yeah, planning occurred by association between risk in the future and decision, our decision in the past.

Here we model that aspect using delayed modulation of cyanotic plasticity mediated by some neural modulator or neurotransmitters.

So this is the model.

This is the model.

This is model.

This is model.

This is model as a.

This is model as a.

The product of the risk factor and the heavy product.

is known to be known all states, possible actions, and sensory inputs for active inference?

Well, you mean if you seek the exact solution, exact optimal solution, then maybe more information would help you to find that.

But under some model, general model, under some ideal assumption, then all variables are not necessary to achieve the optimal solution.

I'm not sure if I correctly answered your point.

So, just to restate it, of course, knowing all the states' possible actions and sensory inputs, it's not a bad thing.

Worst case, there's some computational complexity tradeoffs, but the problem becomes fully statable.

But I think ML Dawn is asking about cases where you don't know all of the state spaces or potentially even the dimensionality or the semantics of hidden states, active states, sensory inputs, and why not even add cognitive states?

So, in not just partially observed but partially known state spaces, how are these addressed in neural networks, and how does active inference help us think about it?

Okay.

I think the question is about how can we separate those states, sensory function, internal, external?

How can we separate not just in principle have these states be separated, but deal with the fact that some of these states we might have good knowledge on, and some states like the hidden states we might not even know?

Like we don't know the dimensionality of the cause vector in the world.

I see.

For example, in terms of dimensionality, there is a statistical technique to estimate the dimensionality.

For example, various information criteria like Akaike information criteria, based information criteria, or then try to infer or estimate plausible dimensionality about the environmental hidden states.

So, there is an analogy with those information criteria and the variation of the free-energy minimization.

So, with variation of the free-energy minimization, we can identify the plausible model structure, which in principle involves the dimensionality aspect.

In terms of neural network, in this paper we don't carefully consider about the dimensionality optimization, because we first define the number of neurons and don't change during training.

But in principle, we can consider the change in the number of neurons, which is associated with the neurogenesis, adult neurogenesis or development during the developmental stage.

That would be an important extension of this direction.

That's very interesting.

Here's a remark.

Well, one note is equation one summarizes a lot of what you've been describing.

There's a parallelism or a concordance being drawn between the loss function of neural networks and the variational free energy of the parameterized model there.

So, to come back to these processes that influence learning, which we could think of as the neural network becoming more fit from a loss function perspective or the variational Bayesian partially observable Markov decision process entity generative model becoming better at doing what it does.

So, there's the firing rate on the neural network side, the synaptic plasticity at a slower timescale, which we discussed a little earlier.

And then now there's a third timescale with the birth and death of new cells and maybe even new layers.

And that kind of multi-scale temporal structuring is not intrinsic to the Bayes graph.

To represent multiple nested timescales in a Bayes graph in the active inference literature, it's more common to make a hierarchically nested model.

Right.

And just say that the time handling on one level is happening more rapidly with respect to clock time than deeper nested slower models.

Whereas the neural formulation allows us to deal with multiple ongoing timescales without appealing to hierarchical nesting, which is a very important feature.

Well, yeah, both, both direction will be possible.

So, without hierarchical modeling or with hierarchical modeling.

So even with hierarchical modeling, the optimization of dimensionality should be possible or would be possible.

But so in other direction, so we can consider that a population of neural models.

So one has a single layer, another has two layers, three layers, four layers, and consider the probability or plausibility of network architecture.

So one has a single layer, and the architecture associated with the performance or cost function minimization under a particular environment, which is in principle, have the same computational architecture with the hierarchical Bayesian model.

So, very interesting.

Yes, perhaps I overgeneralized or speculated because I thought about how one could have a 100 time step POMDP that also performs multi-scale behavior, potentially extremely wastefully.

But at least it could in principle.

And similarly, within a neuron, there could be another neural network or some other structure approximated by that.

So, they almost both enable hierarchical and non-hierarchical modeling, as you described, but in very different ways that lead to very different implementations.

Mm-hmm.

Yes.

I think this brings us to the topic of forward and reverse engineering.

So you talked a lot about reverse engineering.

Yeah.

What is reverse engineering and what is forward engineering?

And what has been done in these areas of engineering?

Okay.

I'm not an expert of reverse engineering in the broad sense, but I believe that reverse here means a characterization of the blueprint of some device or machine.

From data observable information like activity or action behavior of some agent.

Right.

So, goal is identification of the blueprint.

And crucially, here blueprint corresponds to the generative model.

Because once we define generative model, we can derive variational free energy, algorithm, learning algorithm, inference algorithm, and any behavior of agent.

So, here reverse means that we first observe some activity of agent and its mechanism is still unknown for us.

But we can estimate its mechanism using that activity by identifying the most plausible journey model, which can minimize some cost function or risk function.

When we fit the data to the model.

So, on the other hand, forward engineering would be a more mainstream way.

We first define model of blueprint, a genetic model, then derive everything, including variational free energy functional running inference algorithms, behavior action, selection algorithm.

So, by reverse engineering neural networks, we're observing some already parameterized neural network, and then fitting a POMDP to it.

To what extent is it possible to take a given POMDP and create a neural network that performs that inference?

Okay.

In this paper, or in the following paper, what we consider is a strategy that we should use.

We first fit empirical data, whether for neural response data, to a biological plausible canonical neural network model, which is similar to a conventional model fitting approach.

We have differential equation data, where we have differential equation data, and fit data to the differential equation to explain the behavior with the minimal prediction.

So, now, a virtue of this framework we established is that we can naturally transform such neural network architecture with the well-known partially observable macro-condition process architecture, because for any kind of canonical neural network, there is a cost function.

So, we derive cost function from neural activity equation, which is the opposite with the conventional way we first define cost function derived algorithm.

And then, we use the formal equivalence between neural network cost function and variational free energy.

And so, now, transform the neural network architecture to a Bayesian model architecture.

And once we characterize variational free energy, there should be some generative model that define that variational free energy functional.

So, in particular, in this example, canonical neural network, in this example, canonical neural network, nicely correspond to a class of well-known partially observed macro-condition process.

So, by using this procedure, we identify a plausible home-condition architecture which corresponds to observed neural activity data.

So, after all those transformations, first, the measurements of neurons using that data to fit the neural network, and then, by virtue of the relationship known to neural network,

wow well let's stay on this last point so after all those transformations first the measurements

of neurons using that data to fit the neural network and then by virtue of the relationships

unpacked in the paper transforming the neural network in the left side of figure one into
a particular form of the pomdp so first what are the constraints on that form of the pomdp
is this a little corner of model space or what are the space of acceptable pomdps
that totally depends on what kind of neural network model you are considering so for example
in this paper we discuss about a particular class of pomdp that in which each state takes either
0 or 1 so it's very restricted compared to the general form of pomdp but we have we consider
a factorization so in the sense that although each a but we consider a vector of observation
a vector of hidden state where each a element uh correspond to uh once one you know single one
whole vector but as an entire state it can ex represent a high dimensional uh discrete state space
and this architecture nicely correspond to neural network uh architecture because usually each neuron takes
either 0 over 1 or some value continuous variable between 0 and 1 so uh we use this association to
characterize a particular home dp which correspond to neural networks and this follows uh a particular uh
mean field approximation approximation or approximation in generative model because uh we associate a
uh posterior belief in this particular form of the neural network with the neural activity which means that uh posterior of action
also have a factorization architecture in the sense that we don't uh
we don't uh fully consider about the the the second order statistics between
the neural one's activity and neural activity which is outside of this formalism so each
each neuron neuron neuron each neuron's activity corresponds to uh uh posterior expectation about a particular uh element
uh of the state uh of the state and we don't consider the the the the joint probability joint
posterior probability of all states
so although this is a limit limitation of uh uh when we see this as a uh class of uh see as a bayesian
inference but uh otherwise for example we can consider any recurrent network architecture uh which
correspond to a state of transition matrix and uh it would be possible to extend this architecture to
hierarchical structure in the sense that it is straightforward to consider a tree structure or any kind of
uh hierarchical structure by assuming that some neuron connect to other neuron but not connect to
other neuron so this is same as considering the hierarchical structure in general
the hierarchical structure in the hierarchical structure in the hierarchical structure in the hierarchical structure
i think that's very interesting
it's commonly remarked in
the base graphs
that they represent the connections amongst random variables
variables and there's a relationship between their computability and their sparsity the sparsity
structure as in which variables do or do not influence each other
makes the problem tractable through factorization and just kind of conceptually like if every one of
a thousand variables or an unknown number of large variables if it was all by all the number of parameters to
fit on that connectivity matrix would be very high so statistical power would be very low for any given

edge whereas the more and more constrained you make the connectivity of the variables the more statistical power you have to resolve or kind of spend on fitting those edges like in a structural equation

but you might be losing sight of the unknown unknowns by constraining yourself to a very

limited or fallacious topology of the variables so there's this kind of structure learning

statistical inference question in the base graphs then on the neural side from the biological

general much of neuroscience is about understanding how the firing rate connectivity patterns and other

factors how the structure of those neural systems and their function like form and function enable

adequate inference and inference on action so it's like in both of those areas or really like in

neural network artificial and neural networks and in variational bays the discussion is about how

the structure and the fine-tuning work together to generate function and about some of the statistical or

biological challenges of balancing different needs while also constraining the cost

the cost in terms of materials and biometabolism so it's a very rich intersection

that is being explored here

if these models can really be moving back and forth

moving back and forth like there's some imprints of the model that is implementation independent or like

some interlingua or some semantics or compatibility i don't really know i mean that's something we can

explore is like what is it that is such that one could forward engineer and then reverse engineer

and have like kind of an expectation maximization between these two areas so what is it that's being sought

uh yes the important point so yeah for example about you of this relationship is that we can use the knowledge of

uh vision influence to explain your activity and neural dynamics which is crucial because uh people often say that uh

characterization character character characterizing your dynamics is uh not straightforward uh we we may

we may obtain some solution on your network dynamics but the the meaning of that dynamics in terms of

the functional aspect is very unclear we don't know the meaning of connected

e strength is uh matrices and what is the meaning of uh the threshold factor so on so on those are uh

drives as the drive from the you know the modeling the physiological phenomena but uh it is not necessary to

to have a clear linkage to uh functional explanation so explanation of function of the brain but once we transform

or translate these dynamics into a bayesian influence then we can explain every functional aspect of the the

neural network dynamics architecture in terms of uh well established bayesian influence under a particular class of

bayesian modeling in this case pomdp model

so now uh it turns out that uh synaptic strengths correspond to a matrix b matrix which are very established

the clear meaning so yeah this is useful to explain

the neural neural neural synaptic uh property uh in terms of uh well established statistics

uh also for the people in the active inference site you did it would be helpful to understand the neural

neural neural synaptic uh about a particular uh active inference modeling model so uh so i think it's related to forward modeling but uh finally to discuss with the discuss about the vertical

substrate of the substrate of that forward model we need to address the neural network

architecture uh the substrate property so uh in that case uh we can transform a particular

form dp invasion modeling uh to a neural network architecture using this relationship and then get a

prediction about the uh prediction about the substrate so if we have this bayesian model

this this this particular quantity in this model should be

would be

would be possible

oh it's all good can you just repeat the last like 20 seconds uh yes so uh in the last part i mentioned

about for so first we define the Bayesian model and then can predict what is the neuronal substrate

that corresponds to that particular Bayesian model so this will be useful to so identify the

biological you know quantities that correspond to a quantity in Bayesian class yeah

wow well

there's a lot there it makes me think about the differences of implementation and heuristics

in the computational setting which is often in the extreme disembodied and the biological setting

which is in the extreme entirely embodied and for a given generative model

the kinds of computational heuristics that can be applied include a whole host of different strategies

ranging from sampling to tree exploration and branching to parallel paralyzing the data

architecture and all these other kinds of disparate strategies and software packages and implementations

but on the biological side what is needed is something that's very simple but also very

inscrutable which is a given pattern of interactions must embody that calculation

so that might mean that it can add three digit numbers but it can't add two digit numbers

under some constraints

but what isn't accessible to that kind of morphological biological or like form and functional computing what's not accessible are the tree branching

right

when we're thinking about making sentient artifacts or benefiting simply from the explainability

across both sides of this figure

yeah

yeah so you now address an important point so honestly uh it is very non-tubial whether

there is a corresponding biological architecture for any given bayesian architecture

i believe it is impossible to design biological architecture to correspond to arbitrary bayesian architecture

so only a limited uh model can be implemented in a logical plausible manner and that that point

is crucial as a characterization of biological network biological brain

brain right

yeah wow well just to kind of touch again on this forward and reverse engineering

for a given pomdp if we're willing to compose it within a certain class which might be quite general still

but some class of pomdp as written on the paper we may be able to have a neural network architecture
that would be very amenable to deep learning low energy computing pre-training
various features and then on the other side for a given artificial neural network that we come across
in the wild or a model of neural dynamics that we fit using a neural network model
so something in a neuroscience laboratory that model can have interpretability corresponding to the variables
of a given pomdp and just to kind of give one more point on how that's going deeper
than for example statistical parametric mapping spm so let's just assume that the neural network we're
dealing with is fit from brain data from some lucky ant right now what would be possible or prior to this line
of work or without this line of work one could fit a neural dynamics model and then do all kinds of analyses
like power analyses on the different frequency spectra and say look at the average firing rate or the
correlation coefficients of firing rate so fit the firing rates and the synaptic plasticities and and
store all that data and then we could just pick a pomdp that we've seen in the literature without any
reference to the neural network and optimize the pomdp and then we could say well it turns out that when
the pomdp um o is high there's increased theta power in this firing pattern so it's like comparing the descriptive
statistics from the neural model to the descriptive summary statistics of the pomdp decision making model
however with this formal connection there is actually an interpretability
to the unobserved neural states which are what are being inferred from the fmri measurement from the
eeg measurements and so on those underlying variables have a specific interpretability
of the umdp in relationship to the structure of the pomdp
right so yeah that's also very interesting important uh aspect so as you say it's uh what you said is uh
uh i think conventional more conventional uh strategy and it is also formerly related to model comparison
aspect so we usually think various major modeling and identify or select what is the you know best uh
model to explain the given data and this reverse engineering idea involves such a model comparison
aspect in the sense that uh we we try to find the model with the best best explainability which should be
have the identical functional right so so we directly addresses the the the exact
same uh cost function architecture using the transformation natural transformation so uh it should be up to
explain the neural data in the bayesian sense
yeah one can imagine how that would transform the way that
the current neural imaging studies and technologies describe what it is about the measurement that provides
information about the cognitive model so to give another related example let's just say a person was wearing
a eeg headset and a previous study had shown that increased
a band band activity was associated with this behavior that's comparing a descriptive statistic
of the observations of the sensor and correlating the summarized observable to some other variable
variable like anxiety or performance on a behavior in contrast an unobserved variable
in this setting the actual underlying neural state is being correlated to some semantic
the specific generative model component so it's no longer necessarily that any
single frequency band would be associated more or less with a given outcome but it's actually some hidden

state

the variability which gains the interpretability across this transformation which is a subtle

point but it speaks to how broadly the equivalence would reinterpret

empirical neural imaging results as well as a variety of artificial neural network

experiments and diagnostics where people do lesion studies and double knockouts on artificial neural networks

networks so anywhere where somebody with awareness sees that a neural network artificial or biological

is having summary features described and correlated to something that's more semantic

in a quest for meaning may now have a different approach

a model that involves formalizing the model explicitly in terms of unobserved hidden states states with a cost function

akin to a variational free energy minimizing risk bounding surprise on the unobservables

so even though the unobservables were modeled in a sense in the other conventional strategy

like neural activity is a variable in fmri experiments it's underlying the bold signal

the other one is a variable in fmri experiments it's underlying the model yet this formalism concordance

is a more coherent and powerful connection

yeah yeah uh i i i believe so so uh you now address is very important point so

so first to to to address that so we need to clarify about the what is a program uh considered here so this is a a problem so

called meta-veysian uh problem in the sense that researchers try to infer or estimate uh neural activity

a brain activity which infer the external world uh dynamics right so neuron or brain infer environment and we

researcher in for brain activity so there are the two-step of vision influence uh processes so this sort of

meta-veysian uh problem is quite uh tricky uh intractable uh because uh sometimes uh so

probability uh so sometimes random variable uh becomes uh posterior really uh about other aspect so uh i i think

there is some established approach about meta-veysian but uh this is this paper provides some alternative

in the sense that we separate uh two problem uh by by saying that here uh what we infer is a simply

neural network model or neural network dynamics which is shown in the right left hand side of this

this figure so we we fit the data to conventional neural network model which is a simple differential

equation but thanks to this formal equivalence between neural network dynamics and the pomdp uh behavior

then we can transform the the resulting neural network architecture or dynamics into the

the the bayesian inference in some sense post hoc manner so we we nicely avoid the directly addressing the

meta-veysian problem but obtain the same kind of solution in that sense yes uh with combining with

uh brain activity recording like eeg or imaging uh yeah we can we can design we can we can uh estimate a

plausible neural network model uh in the left hand side and we can uh transform that to uh pomdp in the

right hand side

right awesome i'm going to show an image and ask a question from dave in the chat so dave made this image

it's the right side of figure one that we've just been looking at with the variational bayesian formation and he wrote the arc shown as impinging on the s self arc

is this intentional if so it could represent tuning or modulation of the feedback of s into itself

so do you have a thought on this uh it's intentional yes uh i think it's uh related to the usual

uh formulation of uh formulation of uh formdp architecture and the active inference uh context in the sense that

you know our decision or policy a in the usual setting modify uh the state transition matrix uh b matrix

uh right so uh here uh delta is an alternative of policy uh of agent so basically the director indicates a

state transition matrix under a particular decision uh which agent means so in that sense uh what the agent

changes changes is a state transition matrix not the state itself directly that's why we use this this

illustration

awesome very subtle but important point which is when we look at the classical pomdp

formulation so here we'll look at the version shown in figure two i'll just bring just figure two in

um could you describe what you just did about the role of the b matrix in

influencing how hidden states change and how that is where our policies have impact and also please how

do the top and the bottom of figure two differ

um okay so in the in the usual formulation uh under uh active influence uh with pom dp structure so we first

consider uh uh uh the the the the prior preference and uh depending on the prior preference we uh compute

the the expect free energy and its minimization provides the policy and the policy modulate uh

uh does up state transition so now in the upper layer uh we instead use the

filter which is the action of the agent so here action or decision

uh was made uh for each time step so that uh unlike the computational formulation uh we we have a sequence

of uh uh uh uh delta and for each time step delta modulates the state transition matrix b so b is the matrix that uh transform

uh s hidden state in the previous step to the uh s in the current time step

uh and uh it's moderation indicate that uh under a specific uh decision rule uh for example

uh if this s is the uh the virtual environment uh with the goal decision our position move forward

or if we choose the no goal uh decision then it's unchanged so such a uh modulation of state transition was made

uh by uh by choosing delta and the lower part uh correspond to uh bayesian inference made by um bayesian agent

so basically there is a symmetry between upper part and the lower part because uh we assume that this bayesian

is a uh agent has a plausible uh giant model which uh nicely correspond to uh given environment defined in the above

part uh upper part in this figure but uh one interesting

uh things so asymmetry is that uh to model this particular canonical neural network we don't consider

an arrow or a link from uh delta posterior to s posterior which is in the environment

uh delta modulate uh delta modulate uh s in the next step uh through b matrix uh modulation uh in this particular uh

uh bayesian engine which formerly correspond to canonical neural network uh we we don't consider that uh it is

a response to uh an absence of the uh projection from output layer to the the middle the middle layer

okay

let's

let's

this is from the uh 2020 paper but it shows the neural network architecture the two-layer architecture

so could you um restate the top and the bottom of figure two in the 22 paper and connect it to why it's

important that you're studying two-layer neural network models uh i miss you oh yeah can you just connect um

the uh asymmetry between the top and the bottom on figure two with the two-layer neural network architecture

uh

you said that the asymmetry there's no direct link

between

yeah yeah and so this this is another story so in the the previous paper

in the previous paper so there is only um output rate or or

perceptual layer because uh we basically consider a single layer feed for network so

um

this part is identical to a map from O2 S Australia in the 2022 papers.

Ah.

Okay.

So, on the top of figure 2 is the actual generative process.

It's the true structure of causation in the environment, which is to say that actions, delta, actually influence how states change through time via B delta.

Right.

The generative process through the A matrix emits observations, sequences of observations.

And here, on the bottom, with a mirrored structure, is the generative model of the entity.

So, what's the relevance of the arrows and the more Forney factor graph structure on the bottom?

So, the arrow indicates the influence.

So, it's a flow of the information in the sense that to calculate S in the step 2, we use the information of step 2's observation and step 1's posterior expectation of our hidden state.

So, those two determine the S2's expectation.

Usually, in the phonograph, we consider a retrospective arrow.

So, in the sense that S3 also affects the S2 inference.

But this corresponds to a Bayesian smoother.

In the sense that we update every time step simultaneously to better inference.

However, what we consider here is a more filtering approach.

In the sense that for each step, we compute the latest hidden state.

And then we don't change any other state in the past.

So, that's why we don't consider the arrow from future to the past.

Awesome.

Awesome.

Yeah, just to highlight that.

In the Bayesian smoothing approach, it's kind of like fitting a spline.

Because it takes the whole time series and it fits the smoothest possible line.

Or the line whose smoothness is on the AIC-BIC frontier.

But here, on the bottom, with the almost pseudocode implementation provided by the Forney Factor Graph,

which was demonstrated to be equivalent with the Bayesian Graph in the 2017 work with Friston, Par, and DeVries.

This architecture is reflecting a filtering scheme like a Kalman filter or just generalized Bayesian filtering through time where estimates are being carried forward and changed time point to time point.

Such that the decision rules or the updates, perhaps more accurately, are defined between time points.

And the total time series does not have to be loaded into memory or remembered at once.

And then the Bayesian filtering approach has the asymmetry with a different consideration of action.

So, why again is it that action is considered differently in the Bayesian filtering approach on the bottom of the generative model than the consideration of action in the generative process?

Yes.

Yes.

Yes.

This, that corresponds to a lack of connection from Y to X in the figure 1.

Or probably the figure 4 is helpful to that relationship.

So, yeah.

So, yeah.

This is an example network architecture comprising input layer, middle layer, output layer.

What we consider is the information flow from sensory to middle layer.

And middle layer have self-connection, recurrent connection.

And middle layer project to output layer.

So, yeah.

So, there is no connection from all output layer to middle layer.

Right?

So, that's why we don't consider the link from delta in the bottom layer of the figure 2 to S posterior.

So, this is different from true generative process in the environment.

This is a kind of simplification.

So, because our purpose is identifying the plausible Bayesian model which corresponds to this type of true layer neural network, canonical network.

So, in other words, this canonical neural network uses an approximation about that point or, yeah, use a limited form of POMDP scheme.

Thanks.

So, could you describe W, V, K, and gamma?

Just what is the biological or functional interpretation of those variables?

And what brain regions or what processes or pathologies do they map to?

So, basically W, V, K are synaptic strengths in the form of matrix.

So, they represent an object connection in the different layer or different architecture.

In the sense that W means forward connectivity from sensory layer to middle layer.

K corresponds to recurrent network, recurrent connectivity.

And V corresponds to projection from middle layer to output layer.

So, in this paper we don't discuss the relation to brain anatomy in detail.

But one can consider an analogy.

For example, X corresponds to cerebral cortex activity.

And Y corresponds to cerebral.

In the sense that it determines the action.

So, it is considered that in the cerebral there is a signal that represents choice decision.

For example, goal decision or no-goal decision made in cerebral.

It's analogous to this particular architecture.

On the other hand, in the cerebral cortex, we compute the sensory information to generate some inference prediction.

And perhaps planning, so on.

Which is computed by this recurrent network in this particular modeling.

Although we don't separate brain reason in detail.

But this recurrent network is sufficient to.

This class of recurrent network is sufficient to characterize any type of architectures.

In the sense that we can design any.

You know, you know, higher curve.

Or so, inter, so mutually connected architecture.

Using a generic class of recurrent network.

By changing weight matrices.

ail ail

describing how those sensory inputs,

either in one step or composably in multiple steps,

become processed to these X representations

of hidden external causes, inferred external states.

And so these are the states that have that sigma relationship

and a generalized synchrony with external states.
The sigma and the generalized synchrony are not discussed in your paper,
but it connects to other work.
And the recurrent connections are facilitating attention
or weighting of the stimuli.
This is the recurrent learning loop and the relationship of the A
between observations and hidden state estimates.
And then a different kind of modulation comes into play
between the hidden state estimate of the internal state
and the action selection.
So what is gamma corresponding to?
And why is the gamma modulation between layers two and three
differing functionally from the K synaptic modulation of one and two?
Yeah.
So K matrix basically formally correspond to B matrix
in the Bayesian formulation.
So it's located in the information about the prediction, right?
Our knowledge or our expectation about the next state
based on the previous state.
On the other hand,
the role of gamma is quite different from such a computation.
Gamma basically means risk factor, risk function,
which is, yeah, in principle,
we can use arbitrary risk function.
So this is a part of generality model we designed.
And the role of risk function in generality model formulation
is a alternation of form of generality model,
depending on the value of gamma,
which enables a post-dictive or retrospective modulation
modulation of evaluation and task decisions given an outcome in the future.
In terms of neural network,
of course,
it corresponds to some neural modulation,
for example,
dopaminergic modulation is famous in the literature,
which modulates the activity and plasticity
of various brain regions.
But we particularly focus on

modulation, dopaminergic,
or any kind of neural modulation
of cyanotic plasticity
in the output layer,
which may correspond to cerebrum.
So you're in the cerebrum,
neural activity or plasticity
modulated by dopaminergic input
from...

It is used as the optimization
of action rule,
decision rule,
or sometimes attention
and various purpose.
Awesome.
Very interesting,
because in some previous papers
and models that we've looked at,
attention is dealt with
as policy selection
on mental states.
So internal action selection
is an action-like variable
describing attention and awareness
and even metacognition.
Right.
And so that connects
the role of dopamine
in motor decision-making,
seen in many dyskinesias,
but also with the role of dopamine
in seemingly
non-motor-based decision-making,
like gambling or investing,
where it doesn't seem
to immediately translate
to a given motor sequence,

yet it has analogous
computational characteristics
and the comorbidities
and the side effects
of different drugs
that affect the dopamine
neurophysiology
are known to have carryover
in terms of, like,
the rigidity or excessivity
of motor
and decision-making aspects.
So it's, like,
interesting that dopamine
has long been understood
to have that parallel role
in attention
as cognitive action
and motor action,
and that was established
empirically through
modifications of dopamine signaling
and also had been modeled
analogously
with computational neuroscience.
and this is providing,
again,
a slightly different interpretation
of that
very well-studied
dopaminergic modulation
of attention and policy.
Yes.
And in addition to that,
I believe another important aspect
is a modulation
of cyanotic prostitivity
by dopamine.

Well,

I can

show you on...

Do you want to show something?

Or...

Yeah.

Can you see this paper?

I...

I...

I...

I sent you

a link

to a chat.

Or...

Okay.

If you can't,

I'll send you a PDF.

Um...

Okay.

Let me see.

I'll get it up now.

All right.

The paper is...

A critical time window

for dopamine actions

on the structural plasticity

of dendritic spines

from 2014

by Yagashita.

So what is interesting

about this paper?

Yeah.

It basically explains

modulation of plasticity

by dopamine,

which is common,

but a crucial point

of this paper
is that
it shows that
it proved that
dopamine adric
input
can modulate
heavy ampersity
even after
heavy ampersity
is established.
So
this paper
shows that
they add
dopamine adric
input
for example
two seconds
after
or
several seconds
after
the
the
the heavier
plasticity
is established.
But
such a
post hoc
modulation
post hoc
introduction
of
dopamine adric
input
is sufficient

to change
the
past
plasticity
which may
be associated
with
the
post hoc
re-evaluation
of our
past
decisions.
so
by
decision
making
we
of course
change
the
plasticity
change
the
weight
matrix
through
plasticity
but
to
evaluate
the goodness
or badness
of our
decision
we need
to see
observe

the future
outcome
right
which is
propagated
by
for example
dopamine
and this
paper
nicely
show
empirically
that
dopamine
actually
can change
the past
this
joint
evaluation
even after
such a
circuit
level
very
very
local
level
peak
level
hmm
so
there's a
short term
window
the critical
time window

that they're
describing
but there's
some
window
yeah
some
window
yeah
by which
dopamine
potentially
unrelated
to the
initial
heavy and
plasticity
events
right
where
secondary
dopamine
signaling
or
not
secondary
just
after
the
initial
fact
potentially
of a
different
valence
or the
same
valence

can
synergize
or
cancel
the
plasticity
formed
in the
moment
exactly
and this
is not
limited
to
dopamine
but
other
neuromodulator
can also
do
this
well
on one
hand
how does
this
change
our
understanding
of
animal
neurophysiology
and then
I guess
on the
other
hand
how does

this
influence
how we
would
design
sentient
artifacts
when
for
both
animals
and
the
sense
agent
artificial
agent
one
important
message
I
believe
is
that
this
tells
us
possible
simple
architecture
to
make
planning
this
is
association
between
past

decision
and
future
reward
or
any
risk
factors
which
is
otherwise
computed
by
computing
forward
prediction
by
iterating
some
computation
right
I believe
this is
a usual
way
to
predict
the future
event
and
then
select
action
but
using
this
this
property

physiological
property
which is
observed
in
actual
experiment
we can
design
we can
imagine
other
simpler
architecture
to make
a planning
so for
both
animals
sensitive
Bayesian
agent
it
provides
an
alternative
explanation
about
the
association
between
our
past
decision
and
the
future
based

on
the
optimization
of
our
decision
to
maximize
reward
or
minimize
risk
well
one
interesting
note
is
we
spoke
earlier
about
the
difference
between
the
Bayesian
smoothing
all at
once
approach
and
the
Bayesian
filtering
step-by-step
approach
now
if

one
had
infinite
knowledge
and
computational
resources
the
Bayesian
smoothing
approach
is the
way to
go
like
you
don't
want
the
decision
rule
for
investment
you
want
to
look
at
the
whole
time
series
past
present
and
future
and
know

the
best
moments
to
have
made
the
trades
I
mean
there's
no
comparison
you're
going
to
do
better
with
the
Bayesian
smoothing
however
it's
just
implausible
computationally
and because
it requires
total
memory
of the
past
and
knowledge
of
the
future

so
that's
what
motivates
the
development
of
Bayesian
filtering
approaches
which
are
tractable
and
calculable
through
time
yet
with
this
time
delayed
modulation
part
of
the
Bayesian
smoothing
strength
comes
back
into
play
it
doesn't
enable
true
anticipation

of future
states
but that's
what the
expected
free
energy
does
however
the
delayed
neuromodulation
allows for
reconsideration
of a
window
of
past
states
and so
in that
way
it
corresponds
to
like a
slightly
deeper
filter
not just
a filter
of a
time step
of one
but a
filter
of like
a rolling

window
of five
or with
some
decay
and you
don't
want that
window
to be
too
big
because
if
the
window
were
ten
minutes
then
too
many
contrasting
stimuli
would get
piled
together
the
dopamine
level
would
just
converge
to a
mean
field
average
but

there's
some
time
decay
or
time
constant
on
the
post hoc
modulation
where
that
neuromodulatory
signal
is
actually
a
parameter
of
interest
and
that's
not
an
infinitely
long
or
infinitely
short
window
but
it's
some
niche
dependent
amount
of

time
and
that's
a
very
interpretable
and
first
principles
interpretation
of
the
computational
role
of
neuromodulators
in a
way that
is also
consistent
with all
these
other
concordances
we've
been
exploring
so
it's
quite
an
interesting
connection
back
I guess
in our
final
minutes

of this
discussion
what are
you
well
maybe
go to
the beginning
at the
end
which I
meant to
ask
earlier
but it's
a good
way that
we can
sort of
close
today
and look
forward
which is
how did
you
come
to
this
line
of
research
specifically
studying
neural
networks
in
this

way
with
Carl
Fristen
and
your
colleagues
so
yes
so
well
so
my
interest
was
the
characterization
of
biological
network
so
my
first
motivation
is to
make
biological
plausible
artificial
intelligence
but to
achieve that
we need
to know
about
biological
brain
or

biological
neural
network
and
then
so
I
in
these
several
years
I
collaborated
with
the
doctor
professor
Carl
Fristen
to
study
about
his
theory
free
energy
principle
of
Influence
and
then
what
my
question
during
that
period
was

so
is
the
free
energy
principle
is
everything
about
the
you know
biologically
plausible
neural
network
or
is there
any
other
aspect
that
can
characterize
the
biological
neural
network
so
it is
non
trivial
it
was
non
trivial
so
that's
why

I
tried
to
start
from
characterizing
neural
network
first
so
our
strategy
is
not
considering
the
way
of
implementing
any
Bayesian
algorithm
as
brain
architecture
but
my
interest
is
rather
characterization
of
a
given
biological
network
in
terms

of
something
some
other
things
one
possible
way
is
of
course
based
in
inference
free
energy
print
active
reference
so
that's
why
I
first
start
from
characterizing
biological
network
and
but
just
defining
neural
network
architecture
is
insufficient

it is
too
it is
not
tractable
it is
far beyond
the
computational
tractability
as the
mathematical
analysis
and we
need
some
assumption
or some
trick
to
increase
the
tractability
and one
day I
came up
with an
idea
that
so
in which
we
consider
that
both
neural
activity
and

plasticity
follow
the
same
cost
function
gradient
this
is
very
much
an
analogy
with
physical
system
like
Lagrangian
formulation
or
Hamiltonian
formulation
so
usually
we
consider
some
energy
landscape
and
design
plausible
trajectory
as the
solution
of some
principle
of

minimum
action
or least
action
so
we
imagine
that
what if
we
apply
such
idea
to
conventional
canonical
neural
network
or
biological
neural
network
to
characterize
their
dynamics
in the
first
principle
manner
that's
the first
motivation
first
step
to come
up
with

this
framework
mark
and
finally
we

noticed
that
it
is
not
easy
to
connect
the
newtonian
dynamics
with
this
type
of
neural
activity
study
because
neural
activity
equation
is not
necessary
to be
a
second
order
differential
equation
but

rather
it is
first
order
and
considering
many
things
then
we
finally
used
a cost
function
proposed
in
the
papers
which
is
not
necessarily
have
a
formal
identity
with
the
so-called
Lagrangian
in the
Newtonian
physics
but
it is
rather
plausible
as

the
rule
or
underlying
mechanism
of
such
type
of
network
awesome
well
it has
been
quite
an
interesting
dot
one
I
really
appreciate
everything
you've
shared
today
is there
anything
else you
want to
add at
this
point
otherwise
we'll
talk
again
yeah

that

that

is

fine

I

already

speak

girl

thank

you

all right

talk to

you

later

bye

yeah

thank

you

very

much

for

a

nice

discussion

!

Thank you.